

Date: 21/12/2024

Course	: B. Pharmacy	Sem:	I
Subject Name	: Pharmaceutical Analysis-I	Subject Code	: BP102T
Max Marks	: 75	Duration	: 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Write the principle and reaction involved in the limit test for chlorides.
- ii) Explain the concept of oxidation and reduction with example.
- iii) Enlist the types of conductometric titrations.
- iv) Define and classify errors.
- v) Write a note on cerimetry.
- vi) Differentiate between co-precipitation and post precipitation.
- vii) Define buffer and enlist types of buffers.
- viii) Write a short note on diazotization titration.
- ix) Describe briefly masking and demasking agents.
- x) Differentiate between Iodimetry and Iodometry.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Explain various titration curves for aqueous acid base titrations.
- ii) Describe methods of end point detection in precipitation titrations.
- iii) Write a note on impurities and sources of impurities in medicinal agents.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) What are primary and secondary standard substances? Give the requirements to be satisfied by a primary standard substance.
- ii) Describe acid base indicator theories.
- iii) Explain the mechanism of complex formation and add a note on metal indicators.
- iv) Define reference electrode. Explain the construction and working of standard hydrogen electrode.
- v) Write a note on properties of solvents used in non-aqueous titration.
- vi) Draw a neat, labelled diagram of Gutzeit apparatus and explain the principle behind Arsenic limit test.

- vii) Explain construction and working of dropping mercury electrode.
- viii) Explain the principle and steps involved in gravimetric analysis.
- ix) Explain preparation and standardization of 1M Sodium Hydroxide solution.

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